KMTC/QP-08/EPS

KENYA MEDICAL TRAINING COLLEGE
FACULTY CLINICAL SCIENCES
DEPARTMENT OF CLINICAL MEDICINE

END OF SEMESTER EXAMINATION
DIPLOMA IN CLINICAL MEDICINE & SURGERY
CLASS: MARCH 2018 YEAR: 2nd SEMESTER: 1ST
PAPER: SURGERY

DATE: 24th July, 2019	**TIME: 2HOURS (9:00AM – 11:00 AM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (20 Multiple Choice Questions)
   - Section 2 (20 Multiple True/ False Questions)
   - Section 3 (4 Short Answer Questions)
   - Section 4 (2 Long Answer Question)
2. Attempt **ALL** Questions
3. Write your **COLLEGE NUMBER** on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. **DO NOT** write on the Question paper
6. Ensure you sign the examination register provided

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SECTION 1: MCQs

1. Which one of the following is an immediate complication of major abdominal operation?
   a) Hypertrophic scar
   b) Haemorrhage
   c) Wound dehiscence
   d) Dumping syndrome
2. One of the drugs below is usually administered pre-operatively
   a) Adrenaline
   b) Penicillin
   c) Atropine
   d) Diclofenac
3. Spreading inflammation of connective tissues following a thorn prick is referred to as
   a) An abscess
   b) Cellulitis
   c) Carbuncle
   d) Boil
4. The following is a general side effect of local anaesthesia
   a) Vasovagal episode
   b) Nerve ablation
   c) Transcutaneous electrical nerve stimulation
   d) Site stimulation and relaxin
5. All the following are indication of burns except
   a) Electrical burns
   b) Burns on flexure sites
   c) All burns in elderly cohorts
   d) Inhalation burns
6. The following is an early complication of fractures
   a) Fat embolism
   b) Avascular necrosis
   c) Mal-union
   d) Myositis ossificans
7. Which one below is not a radiological method of investigating bone conditions
   a) Radioisotope scanning
   b) Magnetic resonant imaging
   c) Nerve conduction studies
   d) Arthrography
8. Surgical cutting of a bone is referred to as
   a) Bone grafting
   b) Osteotomy
   c) Arthrodesis

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d) Arthroplasty

9. The following is a type of fracture displacement
   a) Oblique
   b) Comminutted
   c) Spiral
   d) Angulation

10. A systemic factor which can hinder healing of fractures includes:
   a) Mobility
   b) Inadequate blood supply
   c) Steroid therapy
   d) Bone infection

11. Regarding compartment syndrome
   a) Normal pressure is less than 10mmHg
   b) Volk man's ischaemic contracture is a common complication in fore arm
   c) Fasciotomy is never indicated
   d) Myoglobinuria is uncommon complication in dehydrated patients

12. The type of bone cells useful in remodelling during fracture healing is
   a) Osteoblasts
   b) Osteoclasts
   c) Osteocytes
   d) Osteons

13. An abnormal complete displacement of articulate surfaces after injury is referred to as
   a) Fracture
   b) Subluxation
   c) Dislocation
   d) Deformity

14. Dirty and contaminated wounds include the following except
   a) Incision and drainage wound after an abscess
   b) Herniorrhaphy
   c) Empyema bladder
   d) Faecal peritonitis

15. Modifiable factors affecting wound healing include
   a) Age
   b) Nutrition
   c) Race
   d) Congenital neuropathies

16. DR—ABCDE used in surgical patient management E stands for the following except
   a) Environment
   b) Evaluation
   c) Evacuation

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d) Exposure

17. Dry gangrene occurs when there is
    a) Occlusion of venous returns
    b) Occlusion of arterial supply
    c) Cut wound
    d) Insect sting
18. Prevention of bed sores include the following except
    a) 2 hourly turning
    b) Urethral catheterization
    c) Avoiding wrinkled sheets
    d) Remaining seated in bed all the time
19. 4th type of diabetic foot occurs when there is
    a) Osteomyelitis
    b) Superficial wound
    c) Gangrene of the entire foot
    d) Gangrene of two toes
20. The best fluids to use when treating dehydration in severe burns is
    a) Blood
    b) 5% dextrose
    c) Haemacil
    d) crystalloids

SECTION 2 TRUE/FALSE

1. The following terminologies are correctly defined
    a) Ectomy------surgical reconstruction of body tissues
    b) Ostomy------formation of permanent or semi-permanent opening in the body
    c) Otomy-------cutting into the organ or tissue
    d) Excision-------removal of foreign body in the body
    e) Oscopy--------minimal invasive procedure that allows visualization of tissues
2. Regarding post -operative surgical care
    a) Feed the patient immediately after major operation
    b) Maintain airway
    c) Recommend intravenous fluids if the patient is feeding
    d) Maintain input-output chart
    e) Mobilize patient out of bed immediately he regains consciousness
3. Medical conditions that affect surgery include
    a) Diabetes mellitus
    b) Hypertension
    c) Fractures
    d) Heart failure

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e) Renal failure

4. Concerning tension pneumothorax
   a) Trachea is deviated to the affected side
   b) Has stonny dullness percussion note
   c) Is synonymous with emphysema
   d) Chest tube is inserted in the 2nd intercostal space
   e) May be a complication of flail chest
5. Bloody nipple discharge can be a feature of the following
   a) Paget's disease of the breast
   b) Acute mastitis
   c) Duct papilloma
   d) Fibro adenoma
   e) Breast carcinoma
6. In management of trauma patients, primary survey involves
   a) Ensuring the airway is clear
   b) Making sure the patient is ventilating
   c) Giving intravenous fluids to restore blood volume
   d) Taking relevant radiological investigations
   e) Performing head to toe examination
7. Common sites for application of skeletal traction include
   a) Upper end of tibia
   b) Calcaneus
   c) Skull
   d) Distal radius
   e) Distal femur
8. In fracture healing
   a) Callus stage precedes inflammatory stage
   b) Bone re-canalizes in remodelling stage
   c) Woven bone is transformed into lamellar bone in consolidation stage
   d) Cellular proliferation Occurs in inflammatory stage
   e) Fractures in children take a shorter period to heal than adults
9. The following laboratory investigations can be requested for orthopaedic conditions
   a) Bence jones protein in urine
   b) Full haemogram
   c) Stool for ova and cyst
   d) Blood for grouping and cross matching
   e) Khan test
10. Concerning grading of muscle power
   a) Grade 4 is normal power
   b) In grade 2,the limb moves against gravity

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c) In grade 1,there is flicker of contraction
d) In grade 3,the limb moves when gravity is eliminated
e) In grade 0,there is no movement

11. Criteria for admission of burnt patient include
    a) Type of burn
    b) Percentage of burns
    c) Co-morbidities
    d) Classification of the burn
    e) Site of the burn
12. Answer true or false concerning cellulitis
    a) It is spread of infection along subcutaneous tissue
    b) Streptococcus and staphylococcus are the common causative mico-organisms
    c) Tetanus toxoid is not indicated
    d) No history of trauma commonly
    e) penicillin's are useful in treatment
13. causes of hypovolaemic shock include
    a) haemorrhage
    b) cardiac tamponade
    c) bad news
    d) dehydration
    e) cholera
14. regarding acute post –operative pain relieve
    a) requires an expert team approach
    b) opioid analgesia should be with held
    c) analgesic dosage must be adequate to control pain
    d) not a must to measure pain levels regularly
    e) a combination of analgesics yields good results
15. the following are principles of fracture management
    a) remodelling
    b) rehabilitation
    c) immobilization
    d) reduction
    e) resuscitation
16. complication of fractures include
    a) mal-union
    b) osteomyelitis
    c) avascular necrosis
    d) fat embolism
    e) compartment syndrome
17. keloids

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a) arise de novo
b) appear immediately after burns
c) surgery is a modality of treatment
d) disappear spontaneously
e) corticosteroids play a role in treatment

18. the following are indication of skin grafting
a) infected wounds
b) clean extensive wound
c) diabetic wound
d) deep wounds with extensive tissue destruction
e) wounds with no signs of re epithelisation

19. treatment of orthopaedic disorders include the following
a) rest
b) application of plaster of Paris
c) skeletal traction
d) insertion of dynamic hip screw
e) cervical collar application

20. classification of fractures according to aetiology involves
a) fatigue fractures
b) fractures caused solely by sudden injury
c) pathological fractures
d) fragility fractures
e) stress fractures

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SECTION 3 SHORT ANSWER QUESTIONS

1. Describe the Rule of 9 in estimation of percentage surface area in burns 10 marks
2. Discuss an abscess under the following
   a) Definition 2marks
   b) Clinical features 4marks
   c) Treatment 4marks
3. A) State 6 methods of internal fixation 6 marks
   b) List 4 indications of external fixation 4 marks

4. Define the following terms (2 marks each)

   a) Bone grafting
   b) Bone transfer
   c) Arthrodesis
   d) Arthroplasty
   e) Osteotomy

SECTION 4 LONG ANSWER QUESTION

1. Discuss pre-operative management of a patient scheduled to undergo exploratory laparotomy (20 marks)